Finite Element Method

Nonlinear FEM

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1. Geometrical Nonlinearity

- Numerous different nonlinearities can occur in solid mechanics which are either of **geometrical** or of **physical** nature.
- The treatment of associated problems demands a large bandwidth of methods and algorithms.
- In structural analysis, it is usually sufficient to consider only small deformations and strains.
- However, there are many problems which depict **large displacements or rotations**, such as cables, beams or shells.
- Such problems require a nonlinear theory which includes the geometry in an exact way.
- Some examples which represent different geometrically nonlinear behaviour are discussed in the following.

Geometrical Nonlinearity — Large Displacements of a Rigid Beam

- The first example for geometrically nonlinear behaviour is a rigid beam of length l (see the figure), which is supported by an elastic rotational spring with stiffness c at its left end.
- Equilibrium at the deformed system yields directly

$$F\,l\cos\varphi = c\,\varphi \tag{1}$$

- Eq. (1) relates the force F in a nonlinear way to the beam rotation ϕ.
- The nonlinearity stems from the **change of geometry** in the equilibrium equation. Hence this type of behavior is known as **geometrical nonlinearity**.

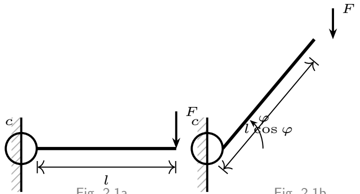


Fig. 2.1a

Fig. 2.1b

- For small rotations ϕ the approximation $\cos\phi \rightarrow 1$ is obtained. With this the **linear solution** $F = c\,\phi\,/\,l$ can be derived from Eq. (1).
- Often so-called **second order theories** are applied to include nonlinear effects. The idea is to describe the nonlinear terms using a Taylor series terminated after the second term. For the presented example:

$$F = \frac{c\,\hat{\varphi}}{l\left(1 - \frac{\hat{\varphi}^2}{2}\right)} \tag{2}$$

- This equation approximates the exact solution up to a rotation of $\pi/3$ very well.
- The solution (2) deviates, however, for larger rotations from the exact solution.

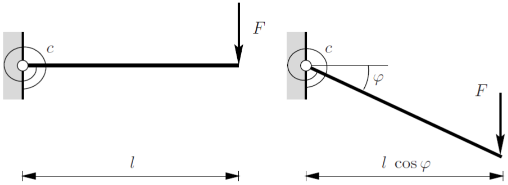


Fig. 2.1a System and loading **Fig. 2.1b** Undeformed system

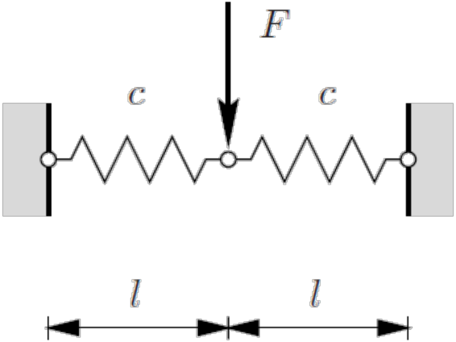
Force $F\,l\,/\,c$ vs. rotation φ (degrees): exact, linear, and second-order theory.

- And
$$\sin\varphi = \frac{w}{l+f}. \tag{4}$$
- Equilibrium follows
$$S_F \sin\varphi = \frac{F}{2}. \tag{5}$$
- Furthermore, the constitutive equation for the spring is assumed to be linear
$$S_F = c\, f \tag{6}$$
where c is the spring stiffness and S_F denotes the force in the spring. Inserting (4) into (5) with (6) yields
$$c\, f\, \frac{w}{l+f} = \frac{F}{2} \tag{7}$$
- This equation can be reformulated using (3) in terms of the vertical displacement w:
$$\frac{w}{l}\left[1 - \frac{1}{\sqrt{1+\left(\frac{w}{l}\right)^2}}\right] = \frac{F}{2\,c\,l} \tag{8}$$
- It is possible to derive a second order theory for small values of w by a Taylor series expansion of the square root:
$$1\Big/\sqrt{1+\left(\frac{w}{l}\right)^2} \approx 1 - \frac{1}{2}\left(\frac{\hat{w}}{l}\right)^2$$

- This approach yields the **cubical polynomial**

$$\frac{\hat{w}}{l}\left[\frac{1}{2}\left(\frac{\hat{w}}{l}\right)^2\right] = \frac{F}{2\,c\,l} \tag{9}$$

- The second order theory approximates the exact solution for $w/l < 0.4$ well.

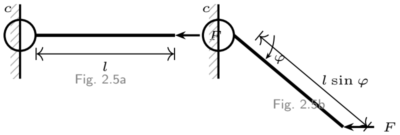


Load–deflection curve $Fl/(2cl)$ vs. displacement w/l: exact theory vs. second order theory.

- The solution of a nonlinear problem is **not always unique**.
- This feature will be discussed by means of the stability problem described in the figure.
- Formulating equilibrium at the deformed system yields

$$F\,l\sin\varphi = c\,\varphi \quad \longrightarrow \quad \frac{\varphi}{\sin\varphi} = \frac{F\,l}{c} \tag{10}$$

- This equation has multiple solutions. The trivial solution is $\phi = 0$, valid for all values of F.
- For $Fl/c > 1$ ($|\phi| \geq |\sin\phi|$), there exist two more solutions. In total, three solutions of Eq. (10) exist for $Fl/c > 1$.
- The point $(Fl/c = 1)$ at which the three different solutions start is known as the **bifurcation point**.



- An approximation of Eq. (10) using the second order theory yields with $\sin\phi \approx \tilde{\phi} - \tilde{\phi}^3/6$ and $1/(1-x) \approx 1 + \tilde{x}$:

$$1 + \frac{\hat{\varphi}^2}{6} = \frac{F\,l}{c} \tag{11}$$

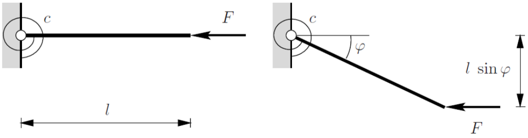


Fig. 2.5a System and loading **Fig. 2.5b** Deformed system

$F/(cl)$ vs. angle φ (degrees): exact vs. second-order theory. Right: graphical solution of $\frac{F\,l}{c}\sin\phi = \phi$.

- In this example, the system consists of two trusses — modelled as springs of length L_0 — under the action of a vertical point force F.
- With the kinematical relations $(h-w)^2 + l^2 = L^2$ and $h^2 + l^2 = L_0^2$ (see Fig. 2.7b), the length change of the spring is:

$$f = L - L_0 = l\left[\sqrt{1+\left(\frac{h-w}{l}\right)^2} - \sqrt{1+\left(\frac{h}{l}\right)^2}\right] \tag{12}$$

- Equilibrium with regard to the deformed system:

$$N\sin(\alpha-\varphi) = -\frac{F}{2} \tag{13}$$

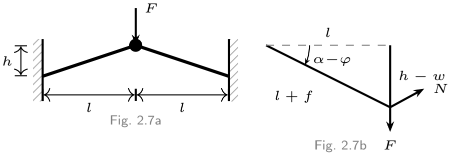


Fig. 2.7a
Fig. 2.7b

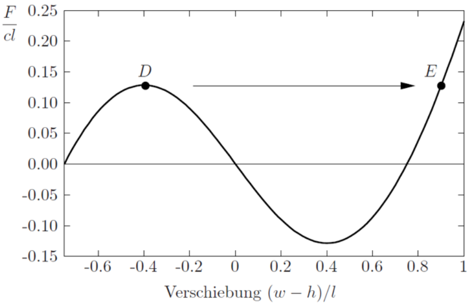
- This relation leads to

$$N\,\frac{h-w}{L} = -\frac{F}{2} \tag{14}$$

- The spring characteristic is linear, hence $N = cf$. Inserting (12) into (14) yields the nonlinear relation between force F and displacement w:

$$c(h-w)\,\frac{L-L_0}{L} = -\frac{F}{2} \implies \frac{w-h}{l}\left[1-\frac{L_0}{l\sqrt{1+\left(\frac{w-h}{l}\right)^2}}\right] - \tag{15}$$

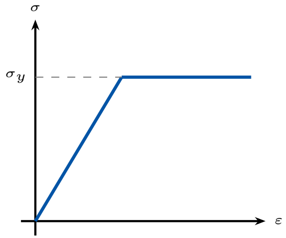
- The associated load–deflection curve is plotted for $L_0/l = 1.25$ in the figure.



Load–deflection curve $F/(cl)$ vs. displacement $(w-h)/l$ $(L_0/l = 1.25)$. D = snap-through point, E = next equilibrium.

- The load increases until point D and decreases afterwards for increasing displacement w.
- The latter situation cannot be reached in case of a static loading. To model this process in a physically correct way, a **dynamic process** should be introduced once point D is passed.
- The system cannot be in static equilibrium until point E is reached for a load larger or equal than the load at D.
- The associated process, in which the system changes from one equilibrium state to another instantaneously, is called **snap-through**. Due to that, point D is called the **snap-through point**.
- Since the solution is not unique at such a point, a snap-through problem is a **stability problem**.
- Depending on the geometry and the loading, snap-through behaviour can be observed in many technical structures like trusses, beams or shells.
- Usually snap-through is connected with a **total failure of a structure**.

- Within the treatment of geometrically nonlinear problems, linear elastic stress–strain relations were used.
- This is a good approximation for many materials but holds under certain assumptions like **small strains**.
- Simple examples show that this is not always true. An elongated rubber band depicts a *greater stiffness with increasing elongation*.
- A metal wire suffers **permanent deformation** under bending due to elasto-plastic material behavior.
- Permanent deformations occur once a limiting stress is exceeded.
- This behavior is also called **plastic flow**, and the limiting stress is known as **yield stress** (see the figure).
- In many technical applications, nonlinear behavior results from a **combination** of geometrical and physical nonlinearities.

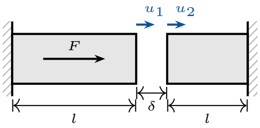


Elasto-plastic stress–strain relation with yield stress σ_y.

- Another but different source for nonlinearities is related to special **boundary constraints**.
- These occur when one body comes into **contact** with another one during a deformation process.
- Here a penetration of one body into the other is ruled out and the contact zone between the two bodies changes depending on the load level.
- Consider two bars with stiffness EA, subjected to a point force F. Both bars are separated by a gap δ.
- We look for the displacement and stress state in the bars on the condition that one bar cannot penetrate the other one.
- This condition leads to an **inequality** for the displacements:

$$u_1 - u_2 \leq \delta \tag{16}$$

- The "$<$" sign holds when bar 1 does not touch bar 2; the "$=$" sign holds at contact.



- In case of $u_1 - u_2 < \delta$, displacement u_2 is zero. Thus, the displacement of bar 1 is

$$u_1 = \frac{F\,l}{EA} \tag{17}$$

- An increase of the force F such that $F > EA\,\delta/l$ leads to contact of bars 1 and 2, then the equation $u_1 - u_2 = \delta$ is valid. Then,

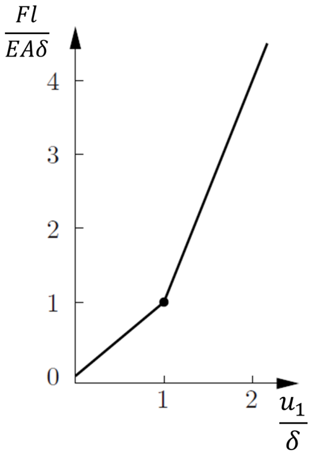
$$u_1 = \frac{F\,l}{EA} - \frac{2N\,l}{EA}, \qquad u_2 = \frac{N\,l}{EA}$$

- For $F > EA\,\delta/l$, it can be shown that

$$\frac{F}{EA} = 3\,\frac{u_1}{l} - \frac{2\delta}{l}$$

- Therefore, the complete force–displacement relation is

$$\frac{F}{EA} = \begin{cases} \dfrac{u_1}{l} & \text{for } u_1 - u_2 < \delta \\[2ex] 3\,\dfrac{u_1}{l} - \dfrac{2\delta}{l} & \text{for } u_1 - u_2 = \delta \end{cases} \tag{18}$$



Force–displacement curve $Fl/(EA\delta)$ vs. u_1/δ. The kink marks the onset of contact between bar 1 and bar 2.

Thanks for your attention!